

EduEase AI – Problem Statement & Agentic AI Solution Overview

IBM watsonx-Powered Course Content Simplification Agent

1. Problem Statement

Educational materials are often written at a level that is difficult for all learners to understand equally. Faculty notes, textbook chapters, and study resources may contain complex explanations, advanced terminology, and dense academic language. Students with different levels of prior knowledge—especially beginners—often struggle to understand key concepts, revise efficiently, and prepare for exams from the same material.

2. Objective of EduEase AI

EduEase AI is designed to solve this challenge by acting as an AI-powered academic assistant that transforms complex course content into learner-friendly study material. The system aims to make academic content easier to understand, more interactive, and more personalized for students with different learning levels.

- Simplify difficult academic content into easy learner-friendly explanations.
- Generate concise summaries for quick revision.
- Create MCQs and exam questions from uploaded notes or pasted text.
- Explain important keywords in simple language.
- Answer student questions directly from the uploaded academic content.
- Adapt all outputs according to learner level: Beginner, Intermediate, or Expert.

3. Proposed Agentic AI Solution

EduEase AI uses an Agentic AI-style workflow in which multiple AI task modules work on the same academic content. Instead of acting like a single basic chatbot, the system performs different educational tasks through dedicated functional modules powered by IBM watsonx.ai and IBM Granite models.

- **Content Simplification Module** – rewrites complex academic text into easier explanations.
- **Summary Module** – extracts the main points for quick revision.
- **MCQ Generation Module** – creates multiple-choice questions with answers.
- **Exam Question Module** – generates important descriptive and short-answer questions.
- **Keyword Explanation Module** – identifies difficult terms and explains them clearly.
- **Question Answering Module** – answers learner doubts using the uploaded content as context.

4. Role of Agentic AI in EduEase AI

Agentic AI allows EduEase AI to behave as an intelligent academic assistant rather than a simple one-step text generator. The system processes one study material input and can perform multiple goal-oriented actions such as simplification, summarization, assessment generation, and question answering. It also supports learner-level personalization by adjusting the complexity of outputs for Beginner, Intermediate, and Expert users.

5. Workflow of the System

- 1 User uploads a PDF or pastes academic text into the EduEase AI application.
- 2 The system extracts and preprocesses the content using Python and PDF processing utilities.
- 3 The learner selects a preferred level such as Beginner, Intermediate, or Expert.
- 4 IBM watsonx.ai with IBM Granite processes the content and generates task-specific outputs.
- 5 Students receive simplified explanations, summaries, MCQs, exam questions, keyword explanations, or answers to doubts.

6. Expected Benefits

- Improves understanding of difficult academic content.
- Supports inclusive learning for students with different knowledge levels.
- Saves time in revision and exam preparation.
- Makes study material interactive through quizzes and question answering.
- Provides a scalable AI-based educational support system for future deployment in colleges and EdTech platforms.

7. Conclusion

EduEase AI addresses the Course Content Simplification problem statement by combining IBM watsonx.ai, IBM Granite, and an agentic academic workflow to convert complex study material into personalized learning support. The solution is practical, student-focused, and suitable for future enhancement with Langflow orchestration, multilingual support, and advanced retrieval-based academic assistance.